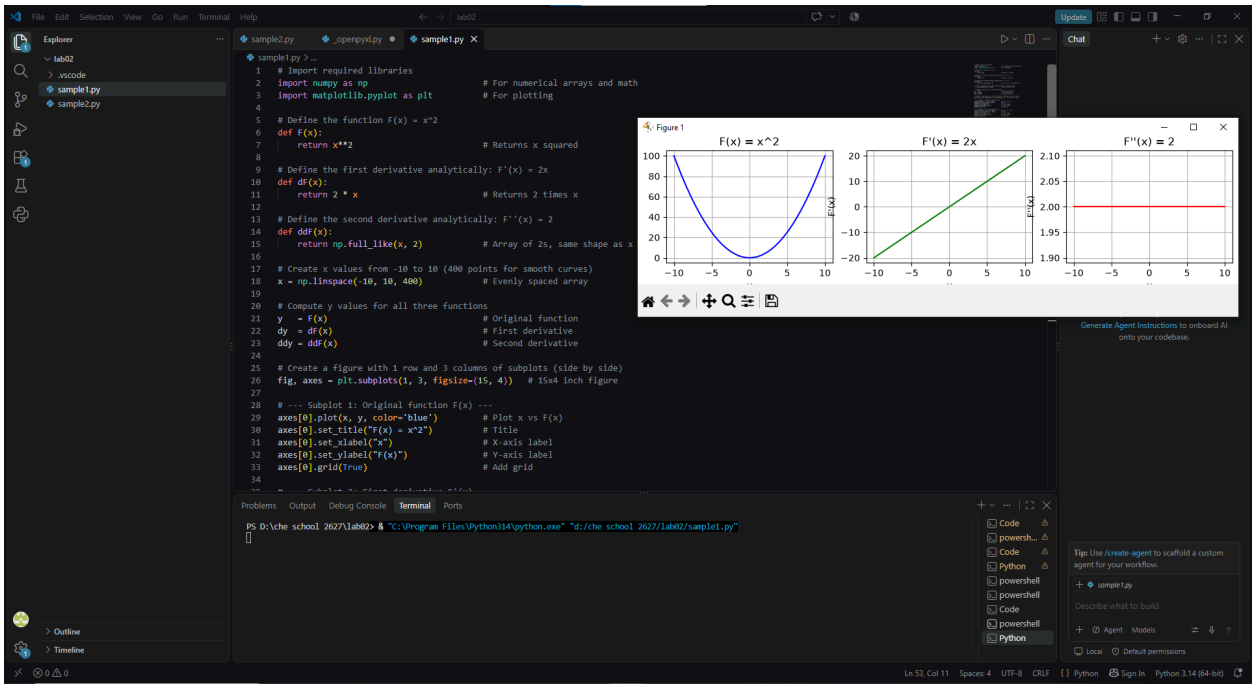
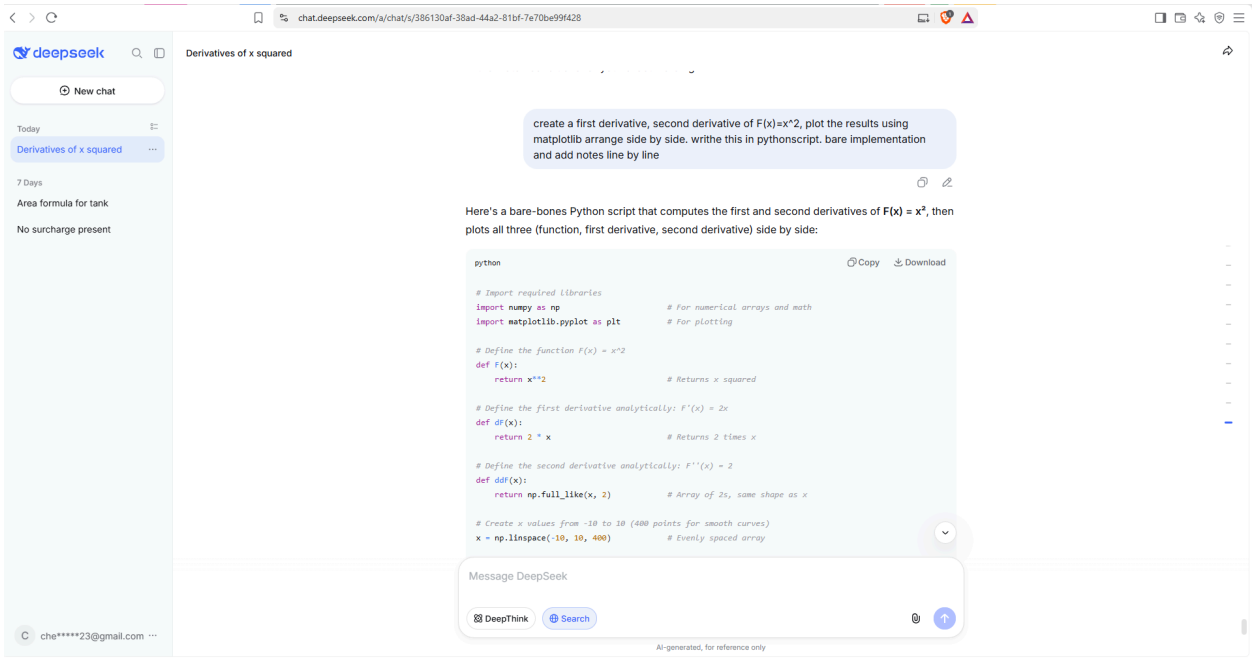
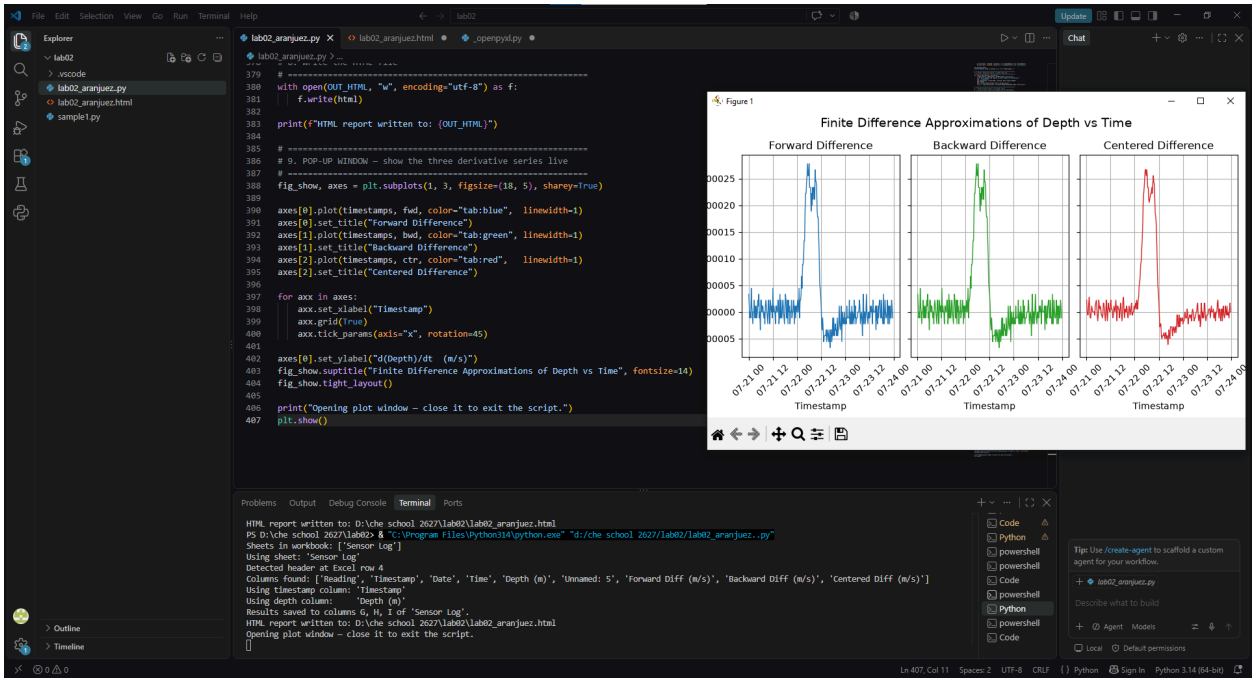
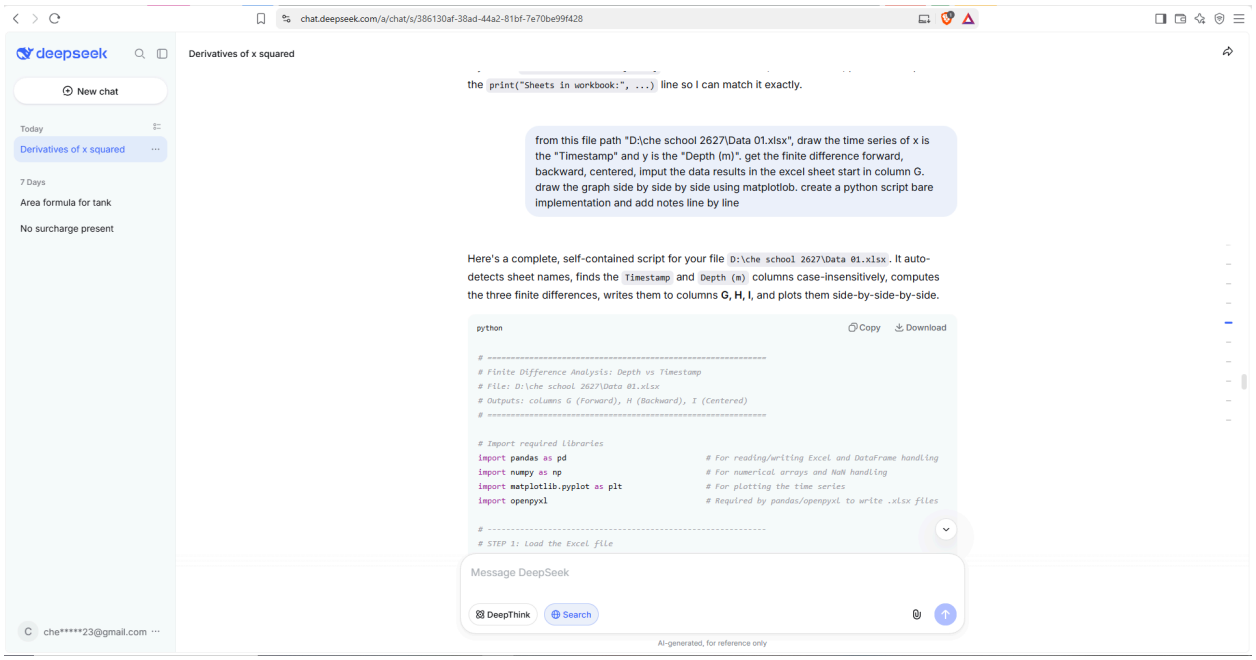
CHE BENEDICT ARANJUEZ TUPM-23-1291

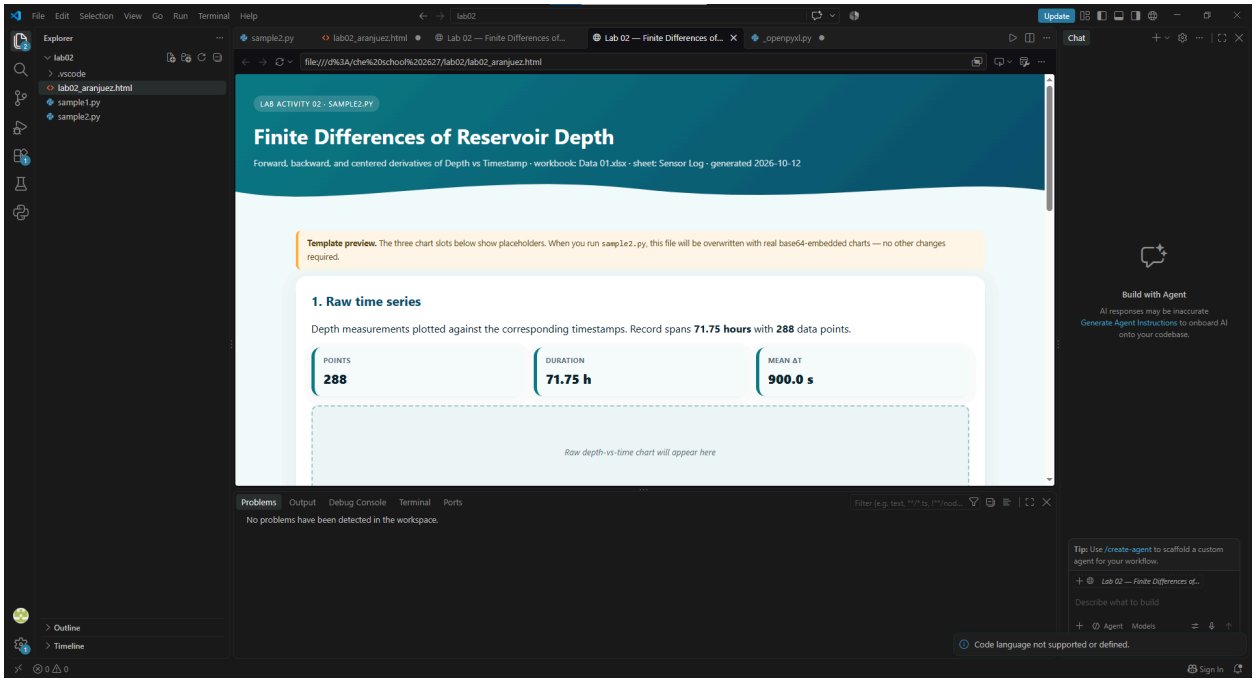


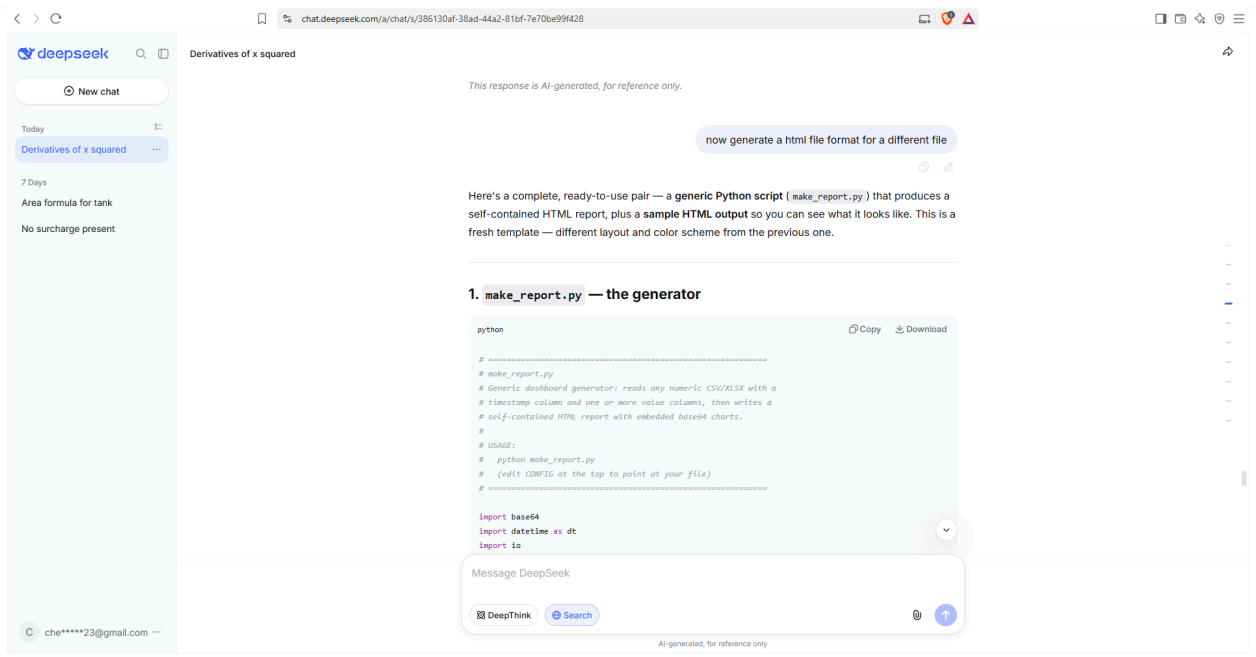



lab02_aranjuez.py
lab02_aranjuez.html
_openpyxl.py
Figure 1
Finite Difference Approximations of Depth vs Time
Forward Difference
Backward Difference
Centered Difference
Timestamp
Problems
Output
Debug Console
Terminal
Ports
HTML report written to: D:\che school 2627\lab02\lab02_aranjuez.html
Sheets in workbook: ['Sensor Log']
Using sheet: 'Sensor Log'
Detected header at Excel row 4
Columns found: ['Reading', 'Timestamp', 'Date', 'Time', 'Depth (m)', 'Unnamed: 5', 'Forward Diff (m/s)', 'Backward Diff (m/s)', 'Centered Diff (m/s)']
Using timestamp column: 'Timestamp'
Using depth column: 'Depth (m)'
Results saved to columns G, H, I of 'Sensor Log'.
HTML report written to: D:\che school 2627\lab02\lab02_aranjuez.html
Opening plot window — close it to exit the script.


deepseek
Derivatives of x squared
New chat
Today
Derivatives of x squared
7 Days
Area formula for tank
No surcharge present
the print("Sheets in workbook:", ...) line so I can match it exactly.
from this file path "D:\che school 2627\Data 01.xlsx", draw the time series of x is the "Timestamp" and y is the "Depth (m)". get the finite difference forward, backward, centered, imput the data results in the excel sheet start in column G. draw the graph side by side by side using matplotlob. create a python script bare implementation and add notes line by line
Here's a complete, self-contained script for your file D:\che school 2627\Data 01.xlsx. It auto-detects sheet names, finds the Timestamp and Depth (m) columns case-insensitively, computes the three finite differences, writes them to columns G, H, I, and plots them side-by-side-by-side.
python
Copy
Download
Finite Difference Analysis: Depth vs Timestamp
File: D:\che school 2627\Data 01.xlsx
Outputs: columns G (Forward), H (Backward), I (Centered)
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import openpyxl
STEP 1: Load the Excel file
Message DeepSeek
DeepThink
Search
AI-generated, for reference only

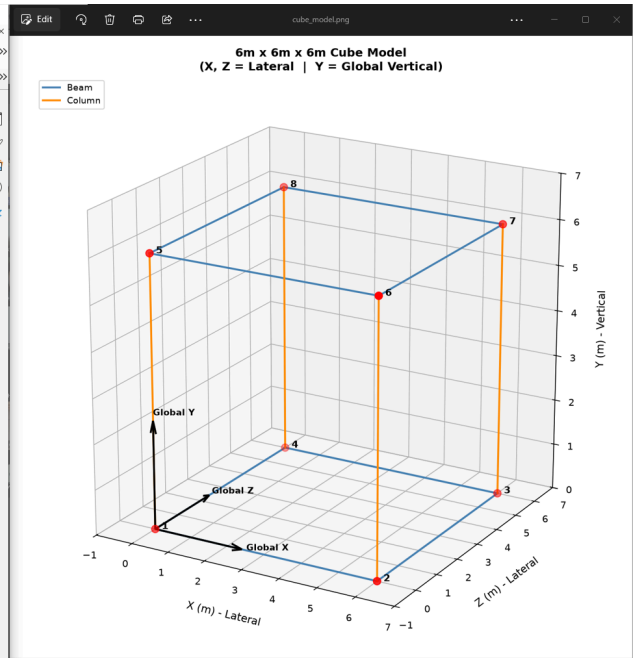
CHE BENEDICT ARANJUEZ TUPM-23-1291





cube_model.xlsx — LibreOffice Calc

Member	i (Start Node)	j (End Node)	Type
1	1	2	Beam
2	2	3	Beam
3	3	4	Beam
4	4	1	Beam
5	5	6	Beam
6	6	7	Beam
7	7	8	Beam
8	8	5	Beam
9	1	5	Column
10	2	6	Column
11	3	7	Column
12	4	8	Column




```
ax.set_xlim(-1, L + 1)
ax.set_ylim(-1, L + 1)
ax.set_zlim(0, L + 1)

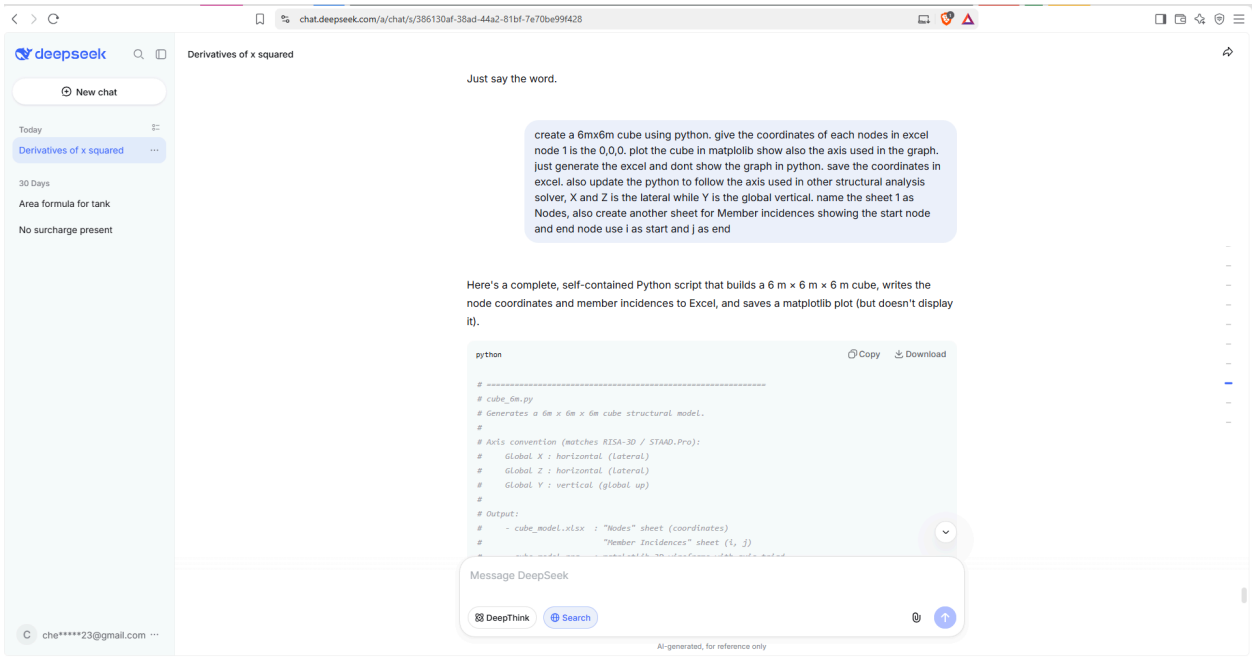
ax.set_box_aspect([1, 1, 1])
ax.view_init(elev=20, azim=-60)

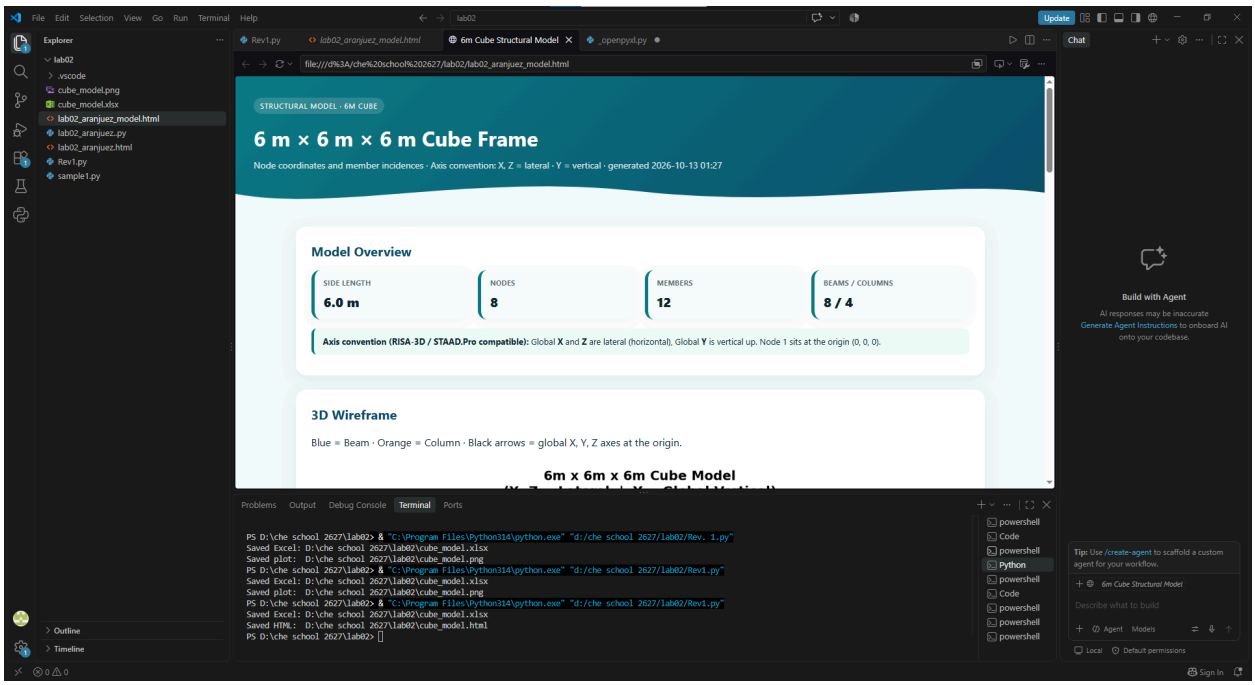
# --- Legend ---
legend_elems = [
    Line2D([0], [0], color="steelblue",  lw=2, label="Beam"),
    Line2D([0], [0], color="darkorange", lw=2, label="Column"),
]
ax.legend(handles=legend_elems, loc="upper left", fontsize=9)

plt.tight_layout()
plt.savefig(PLOT_PATH, dpi=150)
plt.close(fig)
print(f"Saved plot:  {PLOT_PATH}")

# ============================================================
# Done. Note: plt.show() is intentionally NOT called, so no
# graph window will pop up.
# ============================================================
```

```
PS D:\che school 2627\lab02> & "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/sample1.py"
PS D:\che school 2627\lab02> & "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/sample2.py"
Sheets in workbook: ['Sensor Log']
Detected header at Excel row 4
Header: ['Reading', 'Timestamp', 'Date', 'Time', 'Depth (m)', np.float64(nan), 'Forward Diff (m/s)', 'Backward Diff (m/s)', 'Centered Diff (m/s)']
Columns found: ['Reading', 'Timestamp', 'Date', 'Time', 'Depth (m)', 'Unnamed: 5', 'Forward Diff (m/s)', 'Backward Diff (m/s)', 'Centered Diff (m/s)']
Using timestamp column: 'Timestamp'
Using depth column:     'Depth (m)'
Invalid timestamps: 0
Missing depth values: 0
Results written to G, H, I in sheet 'Sensor Log'
PS D:\che school 2627\lab02> "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/sample2.py"
```


chat.deepseek.com/a/chat/s/386130af-38ad-44a2-81bf-7e70be99f428
deepseek
New chat
Today
Derivatives of x squared
30 Days
Area formula for tank
No surcharge present
che*****23@gmail.com
Derivatives of x squared
Just say the word.
create a 6mx6m cube using python. give the coordinates of each nodes in excel node 1 is the 0,0,0. plot the cube in matplolib show also the axis used in the graph. just generate the excel and dont show the graph in python. save the coordinates in excel. also update the python to follow the axis used in other structural analysis solver, X and Z is the lateral while Y is the global vertical. name the sheet 1 as Nodes, also create another sheet for Member incidences showing the start node and end node use i as start and j as end
Here's a complete, self-contained Python script that builds a 6 m × 6 m × 6 m cube, writes the node coordinates and member incidences to Excel, and saves a matplotlib plot (but doesn't display it).
python
Copy
Download
cube_6m.py
Generates a 6m x 6m x 6m cube structural model.
Axis convention (matches RISA-3D / STAAD.Pro):
Global X : horizontal (lateral)
Global Z : horizontal (lateral)
Global Y : vertical (global up)
Output:
- cube_model.xlsx : "Nodes" sheet (coordinates)
"Member Incidences" sheet (i, j)
Message DeepSeek
DeepThink
Search
AI-generated, for reference only


6m Cube Structural Model
STRUCTURAL MODEL · 6M CUBE
6 m × 6 m × 6 m Cube Frame
Node coordinates and member incidences · Axis convention: X, Z = lateral · Y = vertical · generated 2026-10-13 01:27
Model Overview
SIDE LENGTH
6.0 m
NODES
8
MEMBERS
12
BEAMS / COLUMNS
8 / 4
Axis convention (RISA-3D / STAAD.Pro compatible): Global X and Z are lateral (horizontal), Global Y is vertical up. Node 1 sits at the origin (0, 0, 0).
3D Wireframe
Blue = Beam · Orange = Column · Black arrows = global X, Y, Z axes at the origin.
6m x 6m x 6m Cube Model
PS D:\che school 2627\lab02> & "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/Rev. 1.py"
Saved Excel: D:\che school 2627\lab02\cube_model.xlsx
Saved plot: D:\che school 2627\lab02\cube_model.png
PS D:\che school 2627\lab02> & "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/Rev1.py"
Saved Excel: D:\che school 2627\lab02\cube_model.xlsx
Saved plot: D:\che school 2627\lab02\cube_model.png
PS D:\che school 2627\lab02> & "C:\Program Files\Python314\python.exe" "d:/che school 2627/lab02/Rev1.py"
Saved Excel: D:\che school 2627\lab02\cube_model.xlsx
Saved HTML: D:\che school 2627\lab02\cube_model.html
PS D:\che school 2627\lab02>
Build with Agent